

MACHINE LEARNING CHAPTER: CultivaSense Crop Recommendation System

1. Introduction

The Machine Learning component of **CultivaSense** is the intelligence layer that transforms raw soil and environmental parameters into **accurate crop recommendations**. Traditional farming relies heavily on intuition, but CultivaSense replaces this guesswork with a **data-driven classification model** trained on more than **2200 soil samples**.

The ML model evaluates **seven critical parameters** - Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall - and predicts the **most suitable crop among 22 varieties**.

This chapter explains the complete ML pipeline used in the system, covering the dataset, preprocessing, model selection, training, evaluation, and deployment.

2. Dataset Description

The dataset consists of **2200+ labeled samples**, each representing real-world soil and climate readings. Every sample contains:

Input Features (7 total)

1. **Nitrogen (N)** - mg/kg
2. **Phosphorus (P)** – mg/kg
3. **Potassium (K)** – mg/kg
4. **Temperature** – °C
5. **Humidity** – %
6. **pH Level** – scale
7. **Rainfall** – mm

Output Label (1 total)

- **Crop Category** (22 possible crops: Rice, Wheat, Mango, Grapes, Coffee, Cotton, Jute, etc.)

The dataset is **tabular, structured, and multi-class**, making it ideal for classical ML algorithms.

3. Data Preprocessing

Raw data cannot be directly used for model training because the features have **different scales**. For example:

- Potassium ranges between 0 - 200
- pH ranges between 0 - 14
- Rainfall ranges between 0 - 300

If scaling is not applied, features with larger numerical values dominate model learning.

Therefore, **Min–Max Scaling** is used.

3.1 Min–Max Scaling Formula

$$\text{scaled_value} = (x - \min) / (\max - \min)$$

After scaling, all features lie between **0 and 1**, ensuring fair contribution.

3.2 Preprocessing Pipeline

1. Handel missing values (if any)
2. Apply Min-Max Scaling on all feature columns.
3. Store scaler as **minmaxscaler.pkl** for later use.

This normalization is essential for maintaining **model stability** and **accuracy consistency** across user inputs.

4. Model Selection: Random Forest Classifier

CultivaSense uses the **Random Forest Classifier**, chosen after comparing multiple algorithms (Naive Bayes, Decision Tree, SVM, XGBoost).

Random Forest performs best for agricultural datasets because:

✓ Handles non-linear relationships

Soil nutrient interactions are complex and non-linear. RF handles this easily.

✓ Resistant to overfitting

Uses **multiple decision trees** instead of one.

✓ High accuracy

Achieves **95% accuracy** in CultivaSense.

✓ Works well with tabular data

Agricultural datasets are structured, and Random Forest excels in these conditions.

4.1 How Random Forest Works

1. Builds many individual decision trees.
2. Each tree gives a crop prediction (e.g., Rice / Jute / Maize).
3. Final output = **majority vote** from all trees.

Example:

Tree1 → Rice

Tree2 → Rice

Tree3 → Maize

Final prediction → **Rice**

This voting mechanism increases reliability.

5. Training and Validation

5.1 Train-Test Split

The dataset is divided into:

- **Training Set:** 80%
- **Validation Set:** 20%

5.2 Cross-Validation

K-Fold Cross Validation ensures the model performs well across different parts of the dataset, confirming **generalization**.

5.3 Training Details

- Algorithm: Random Forest
- Trees (Estimators): 100–200 (optimal)
- Criterion: Gini Impurity
- Depth: Auto-selected based on data patterns

During training, the model learns the patterns connecting soil attributes to correct crop labels.

6. Model Evaluation

CultivaSense evaluates the ML model using:

6.1 Confusion Matrix

The confusion matrix displays the **number of correct vs incorrect predictions** for each crop class. Interpretation:

- **High diagonal values** → Lots of correct predictions
- **Low off-diagonal values** → Very few misclassifications
- Strong class separation confirms that nutrient features are strong predictors

This matrix is essential to demonstrate reliability to evaluators.

6.2 Accuracy Curve (Training vs Validation)

The accuracy graph shows how well the model learned during multiple training cycles:

- Rapid accuracy growth during the first few epochs
- Training accuracy stabilizes $\approx 96\%$
- Validation accuracy stabilizes $\approx 94\text{--}95\%$
- The two curves stay close → **Model is NOT overfitting**

This proves the model performs well on **unseen data** and is reliable for real farmers.

7. Model Workflow (End-to-End Prediction Pipeline)

When a farmer or user enters values into the web app:

Step 1: User Input

N, P, K, Temperature, Humidity, pH, Rainfall

Step 2: Scaling

Values are normalized using the stored **minmaxscaler.pkl**

Step 3: Prediction

Random Forest model (**model.pkl**) predicts the crop.

Step 4: Insight Generation

The system compares user values with ideal crop conditions.

Step 5: Recommendation Output

- Best crop
- Soil insights
- Deficiency alerts
- Nutrient recovery plan

This pipeline ensures real-time, accurate decision support.

8. Deployment of ML Model

The trained ML model and scaler are exported as:

- **model.pkl**
- **minmaxscaler.pkl**

Both files are loaded inside **app.py** using Flask.

The model runs entirely **locally** and produces **instant predictions** without requiring GPUs or cloud infrastructure.

9. Conclusion

The Machine Learning module of CultivaSense successfully transforms soil data into actionable crop recommendations with **95% accuracy**.

Using preprocessing, robust modeling, and real-time deployment, the system:

- Improves crop selection
- Increases yield potential

- Reduces nutrient wastage
- Enables scientific farming for rural communities

This ML framework is scalable, reliable, and can easily integrate advanced features such as weather prediction, IoT sensors, and fertilizer optimization in future versions.